Sean L. Snaider

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EDUCATION

Northeastern University **Boston, Massachusetts**

B.S. in Computer Science, Concentration in Software Engineering *Expected May 2028*

- **GPA:** 3.89/4.00, *Dean's List*

EXPERIENCE

Vertex Pharmaceuticals **Boston, Massachusetts**

Software Developer Co-op Jul 2026 - Dec 2026

- Designed and built a Python/SQL ETL tool migrating ~4,000 contracts (~200 fields each) from Agiloft to Icertis, replacing a manual process and a legacy Microsoft Access workflow; management projects ~$100K/year in savings.
- Made migrations reconfigurable without code changes by driving all ~200 field mappings from declarative Excel config with per-field overrides; iterated on the design with ~10 stakeholders, including the heads of contract management and Icertis.
- Built automated validation tests that check migrated records for value fidelity, referential integrity, and datatype mismatches against the Icertis schema, narrowing manual review to logic-level issues.

Student Government Association, Northeastern University **Boston, Massachusetts**

Software Engineer *Feb 2026 - Present*

- One of 6 engineers on a full-stack attendance and voting app (Next.js, Prisma, PostgreSQL) that replaced spreadsheets for 50+ members across 20+ meetings per semester; built UI features, fixed attendance data-loss bugs, and review and merge teammates' pull requests.
- Fixed a data-loss bug where editing one member's attendance reset every other member's recorded attendance to "pending"; traced it to an update loop that overwrote finalized records instead of skipping them.
- Blocked ineligible members from voting in executive elections by moving vote validation server-side (Next.js API routes, Prisma, schema-checked requests).

Khoury College of Computer Sciences, Northeastern University **Boston, Massachusetts**

Teaching Assistant *Sep 2025 - Apr 2026*

- Led two weekly labs of ~40 students for intro CS courses in Python and Java (850+ combined enrollment), covering OOP, data structures, and recursion; graded assignments and ran review sessions.

PROJECTS

BellyDSP: Open-Source Guitar Amp Simulator - *C++20, JUCE, Python, CMake, GitHub Actions* - Github *Sep 2026 - Present*

- Built and released a free neural-network guitar amp simulator for macOS and Windows; wrote the original Rust implementation by hand, then designed the architecture for an AI-assisted C++20/JUCE port and now develop new features in C++ by hand.
- Enforced real-time safety with GitHub Actions CI tests that fail if the audio thread allocates memory or takes a lock, the two most common causes of audio dropouts.
- Diagnosed and fixed two performance bottlenecks: ran inference only for the selected amp, cutting CPU usage ~4x while keeping amp switches seamless, and replaced per-sample std::tanh with Eigen's vectorized tanh, doubling neural-network throughput.
- Built tone matching that isolates the guitar from a full mix with Demucs and fits the app's own gear to it; gated every change on a 50-song benchmark, reducing matching error 31%.

Guitar Learning Tool - *React, TypeScript, FastAPI, SQLAlchemy, Anthropic API - Github* *Feb 2026*

- Built a browser-based chord detector using an FFT chromagram and benchmarked it on 180 real recordings from the public GuitarSet dataset, correctly identifying root and major/minor quality 73% of the time.
- Generated chord progressions and personalized practice plans through a FastAPI REST API backed by the Anthropic API, using structured outputs to constrain responses to typed schemas the app renders reliably.

Rubik's Cube Solver - *Python, React, TypeScript, FastAPI, MongoDB - Github* *Feb 2026*

- Implemented Kociemba's two-phase algorithm from the original papers behind a FastAPI REST API, solving scrambles in ≤20 moves in under 1s in testing.
- Traced intermittent failed solves to a fixed-orientation assumption by testing scrambles on a physical cube, then normalized every state to a canonical orientation before solving.

SKILLS

Languages: Python, Java, C++, C, TypeScript, Rust, SQL
Frameworks & Data: JUCE, React, Next.js, FastAPI, SQLAlchemy, Prisma, PostgreSQL, MongoDB, Pandas, NumPy
Tools & Infrastructure: Git, GitHub Actions (CI), Docker, CMake, Linux, REST APIs, Anthropic API (structured outputs, tool use)